

LOOPS And Matchcases:

Q1. What is the difference between for and while loops in Python?

- A) for is used mainly for iterating over an iterable, while while repeats based on a condition
 - B) for can never be nested
 - C) while can iterate over lists but for cannot
 - D) There is no difference
-

Q2. What does the range() function return in Python 3?

- A) A list
 - B) A tuple
 - C) A range object
 - D) A generator
-

Q3. What is the difference between break and continue?

- A) break skips one iteration, continue terminates the loop
 - B) break terminates the loop, continue skips the current iteration
 - C) Both terminate the loop
 - D) Both skip the current iteration
-

Q4. In nested loops, what does break normally terminate?

- A) All loops
 - B) The outermost loop
 - C) The nearest enclosing loop
 - D) The entire Python program
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Q5. What is the purpose of case _ in match-case?

- A) It matches only _
 - B) It works as a wildcard/default case
 - C) It causes an error
 - D) It matches only strings
-

Q6. What is a guard in Python match-case?

- A) A condition written after a pattern using if
 - B) A loop inside match
 - C) The _ pattern
 - D) A replacement for break
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Q7. What happens when multiple case patterns can match the same value?

- A) All matching cases execute
 - B) The last matching case executes
 - C) The first matching case executes
 - D) Python randomly selects one
-

Q8. What is the output?

```
for i in range(1, 5):  
    for j in range(i):  
        print(j, end=" ")
```

- A) 0 0 1 0 1 2 0 1 2 3
 - B) 0 1 2 3
 - C) 0 1 0 1 2 0 1 2 3
 - D) 1 2 3 4
-

Q9. What is the output?

```
for i in range(1, 5):  
    if i == 2:  
        continue  
    for j in range(2):  
        print(i, j, end=" ")
```

- A) 1 0 1 1 3 0 3 1 4 0 4 1
 - B) 1 0 2 0 3 0 4 0
 - C) 1 0 1 1 2 0 2 1
 - D) 1 1 3 1 4 1
-

Q10. What is the output?

```
for i in range(3):  
    for j in range(3):  
        if j == 1:  
            break  
        print(i, j, end=" ")
```

- A) 0 0 1 0 2 0
- B) 0 0 0 1 1 0 1 1
- C) 0 1 1 1 2 1
- D) Nothing

Q11. How many times will X be printed?

```
for i in range(1, 5):  
    for j in range(1, 5):  
        if i == j:  
            continue  
        print("X")
```

- A) 4
 - B) 8
 - C) 12**
 - D) 16
-

Q12. What is the output?

```
x = 0  
for i in range(3):  
    for j in range(2):  
        x += 1  
    x += 10  
print(x)
```

- A) 6
 - B) 30
 - C) 36**
 - D) 33
-

Q13. What is the output?

```
for i in range(2, 10, 3):  
    print(i, end=" ")
```

- A) 2 5 8**
- B) 2 5 8 11
- C) 3 6 9
- D) 2 4 6 8

Q14. What is the output?

```
for i in range(1, 6):  
    if i % 2 == 0:
```

```
continue
```

```
if i == 5:
```

```
    break
```

```
print(i, end=" ")
```

A) 1 3

B) 1 3 5

C) 2 4

D) 1 2 3 4

Q15. What is the output?

```
x = 20
```

```
match x:
```

```
    case n:
```

```
        print(n)
```

```
    case 20:
```

```
        print("Twenty")
```

A) 20

B) Twenty

C) 20 Twenty

D) Error

Q16. What is the output?

```
x = 10
```

```
match x:
```

```
    case n if n > 20:
```

```
        print("A")
```

```
    case n if n > 5:
```

```
        print("B")
```

```
    case _:
```

```
        print("C")
```

A) A

B) B

- C) C
 - D) Error
-

Q17. What is the output?

x = 3

match x:

case 1 | 2 | 3:

print("A")

case 3:

print("B")

case _:

print("C")

- A) A
 - B) B
 - C) A B
 - D) C
-

Q18. What is the output?

x = (1, 2)

match x:

case (1, 2):

print("A")

case (1, 3):

print("B")

case _:

print("C")

- A) A
 - B) B
 - C) C
 - D) Error
-

Q19. What is the output?

x = (10, 20, 30)

```
match x:
    case (a, b, c):
        print(a + b + c)
    case _:
        print("Invalid")
```

- A) 30
 - B) 60**
 - C) (10, 20, 30)
 - D) Invalid
-

Q20. What is the output?

```
x = [1, 2, 3, 4]
match x:
    case [a, b, *c]:
        print(a, b, c)
    case _:
        print("Invalid")
```

- A) 1 2 [3, 4]**
 - B) 1 2 3 4
 - C) [1, 2] [3, 4]
 - D) Invalid
-

Q21. What is the output?

```
x = [10, 20, 30, 40]
match x:
    case [10, *rest]:
        print(rest)
    case _:
        print("No")
```

- A) [10]
 - B) [20, 30, 40]**
 - C) 10
 - D) No
-

Q22. What is the output?

```
for i in range(1, 6):
```

```
    match i:
```

```
        case 1 | 3 | 5:
```

```
            print("Odd", end=" ")
```

```
        case 2 | 4:
```

```
            print("Even", end=" ")
```

A) Odd Even Odd Even Odd

B) Even Odd Even Odd Even

C) Odd Odd Odd

D) Error

Q23. What is the output?

```
for i in range(1, 6):
```

```
    match i:
```

```
        case n if n % 2 == 0:
```

```
            continue
```

```
        case _:
```

```
            print(i, end=" ")
```

A) 1 2 3 4 5

B) 2 4

C) 1 3 5

D) Nothing

Q24. What is the output?

```
for i in range(1, 6):
```

```
    match i:
```

```
        case 3:
```

```
            break
```

```
        case _:
```

```
            print(i, end=" ")
```

A) 1 2

B) 1 2 3

C) 3 4 5

D) Error

Q25. What is the output?

```
for i in range(1, 4):
```

```
    match i:
```

```
        case n if n == 2:
```

```
            print("Two")
```

```
        case n:
```

```
            print("Other")
```

A)

Other

Two

Other

B)

Two

Other

Other

C)

Other

Other

Other

D) Error